Muhamad Akhdan Ramadhan

Computer Engineering Technology Student | Software, Networks & Embedded Systems

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Profile

Fifth-semester Computer Engineering Technology student at Sekolah Vokasi IPB University. Hands-on personal and academic work across software, computer networks, and microcontrollers. Builds with AI coding tools while directing implementation, auditing changes, and operating the resulting systems.

Education

IPB University, Sekolah Vokasi | Bogor, Indonesia
Teknologi Rekayasa Komputer (Computer Engineering Technology), D4 applied bachelor's program
Semester 5 | Angkatan 61

Selected personal & academic projects

Multi-agent automation on a 2 GB VPS | Personal project

Built and operate nine Linux/systemd services: an Orchestrator, an engineering Worker, and seven specialist agents, using AI assistance. Telegram control and a central audit queue hold outgoing content if review is unavailable. Added quota-only model fallback through 9Router, a shared RAM gate for video rendering, watchdog monitoring, guarded auto-fix, and automatic rollback. Worker supports FIFO jobs, live status, emergency stop, and search across 917 skills. Trading experiments use code risk gates before AI analysis, with small personal capital; profitability is unproven.

SmartCow monitoring & sprinkler control | Semester 5 / In progress

Developing a web dashboard and local control system with two ESP32 nodes, ESP-NOW, Firebase, six DS18B20 temperature sensors, and four DHT22 humidity sensors. Current logic includes data-freshness checks, temperature/THI thresholds, hysteresis, duration limits, and cooldown. Prototype; cattle-shed field testing remains incomplete.

Account-based boarding-house Wi-Fi | MikroTik coursework

Configured a Wi-Fi system requiring residents to log in with a boarding-house account before connecting, applying network access and authentication concepts.

ESP theft detector | Semester 4

Built a theft-detection prototype using an ESP microcontroller and an ultrasonic sensor.

Serverless utilities & integration tools | Personal projects

Connected personal Android notifications to Cloudflare Workers and D1 via MacroDroid webhooks, with Telegram device-location requests. Other tools include cross-device service handoff and a SQLite-backed chat summarizer.

Technical skills

Project tools: JavaScript/Node.js, Python, SQL, PowerShell, Linux, systemd, REST APIs, LLM APIs, SQLite, Git, Cloudflare Workers/D1, Docker Compose, n8n, browser tools, FFmpeg.
Coursework & academic practice: Cisco networking, MikroTik, ESP/ESP32, Arduino, computer architecture, PLC logic, CX-Programmer, Factory I/O, data processing in RStudio, MATLAB.

Cisco Networking Academy course completion

CCNA: Switching, Routing, and Wireless Essentials | 17 Dec 2025
CCNA: Introduction to Networks | 29 Dec 2024
Course-completion certificates; not a professional CCNA certification.